

Inventory and ordering decisions: a systematic review on research driven through behavioral experiments

Inventory and
ordering
decisions

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Abstract

Purpose – The success of a supply chain is highly reliant on effective inventory and ordering decisions. This paper systematically reviews and analyzes the literature on inventory ordering decisions conducted using behavioral experiments to inform the state-of-the-art.

Design/methodology/approach – This paper presents the first systematic review of this literature. We systematically identify a body of 101 papers from an initial pool of over 12,000.

Findings – Extant literature and industry observations posit that decision makers often deviate from optimal ordering behavior prescribed by the quantitative models. Such deviations are often accompanied by excessive inventory costs and/or lost sales. Understanding how humans make inventory decisions is paramount to minimize the associated consequences. To address this, the field of behavioral operations management has produced a rich body of research on inventory decision-making using behavioral experiments. Our analysis identifies primary research clusters, summarizes key learnings and highlights opportunities for future research in this critical decision-making area.

Practical implications – The findings will have a significant impact on future research on behavioral inventory ordering decisions while informing practitioners to reach better ordering decisions.

Originality/value – Previous systematic reviews have explored behavioral operations broadly or its subdisciplines such as judgmental forecasting. This paper presents a systematic review that specifically investigates the state-of-the-art of inventory ordering decisions using behavioral experiments.

Keywords Inventory and ordering decisions, Behavioral operations management, Supply chain, Bibliometric and network analysis, Systematic literature review

Paper type Literature review

1. Introduction

Effective and efficient flow and storage of products and raw materials is paramount to the success of a supply chain (Cachon and Terwiesch, 2006). Organizations rely on experienced practitioners to make inventory and ordering decisions. Aided by information systems, individuals make a call on how much to order, when to order and what amount to hold in



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inventory (Williams and Tokar, 2008). The effectiveness of these decisions depends on numerous factors such as targeted customer service levels, variability in demand and transit times, inventory on hand and associated ordering and inventory holding costs. Evidence shows that decision makers tend to deviate from what is prescribed by quantitative (analytical and optimization) models (Becker-Peth and Thonemann, 2018), resulting in unnecessary costs and operational inefficiencies. One of the most well-known and highly disruptive results of these deviations from optimal behavior is the bullwhip effect (BWE), which describes the phenomenon of order variance amplification in supply chains. While normative behavior can lead to the BWE (Lee *et al.*, 1997), studies conducted in a controlled laboratory environment provide evidence that the problem persists even when researchers control for these rational causes (Croson and Donohue, 2002, 2003, 2006), demonstrating that irrational decisions by managers can be a significant source of the problem. These experiments serve to highlight the critical impact that human decision makers can have on supply chain efficiency through their ordering activities, and the importance of understanding this behavior.

In response to the growing recognition of understanding human behavior, the field of behavioral operations management (BOM) has emerged (Bendoly *et al.*, 2006; Donohue *et al.*, 2020; Fahimnia *et al.*, 2019). BOM “explores the interaction of human behavior and operational systems and processes” (Bendoly 2015; pp. 9). Within BOM, the topic of ordering decisions has emerged as a major focal area (Becker-Peth and Thonemann, 2018). While normative research has investigated this phenomenon using computational methods, they have been inadequate to fully explain human decision-making (Kremer and Minner, 2008). This has prompted researchers to draw inspiration from other fields of research such as economics, finance, psychology, marketing, sociology, medicine and accounting, which had all embraced research on human behavior prior to the emergence of BOM (Bendoly *et al.*, 2006).

Laboratory experiments tend to be the most popular research method in this context – and for good reason. Lab experiments allow researchers to maintain the high degree of control necessary to isolate and identify relationships between variables of interest and behavior (Bachrach and Bendoly, 2011; Katok, 2018; Siems, 2011). Thus, they are ideally suited for establishing causal connections and examining the effect of behavioral factors (Croson and Donohue, 2002; Sekaran, 1992; Wacker, 1998). As a result, they provide an excellent platform to test economic theories (Bendoly *et al.*, 2006).

Reviews of the existing literature on certain types of behavioral inventory and ordering decisions have been published as chapters in edited books. One of the earliest works focuses exclusively on behavioral experiments focusing on the newsvendor problem (Kremer and Minner, 2008). The recently released *The Handbook of Behavioral Operations* also contains a chapter on behavioral inventory decisions (Becker-Peth and Thonemann, 2018). Becker-Peth *et al.*'s (2018) chapter primarily focuses on the literature of single-period inventory decisions (newsvendor problem) using behavioral experiments or computational methods. While these reviews have contributed to the advancement of the field, it is important to expand beyond single-period ordering decisions given the increasing significance of supply chain decision-making. This paper aims to address this gap by presenting a systematic review and classification of literature on behavioral ordering and inventory decisions that pertain to single-period, multi-period and the beer distribution game contexts for the first time.

Systematic literature reviews are an important contributor in advancing sciences. Given the methodical and reproducible nature of systematic reviews, biases in collating literature can be minimized (Durach *et al.*, 2017). Numerous systematic reviews have contributed to the growth of supply chain management (Arvan *et al.*, 2018; Burgess *et al.*, 2013; Eksoz *et al.*, 2014; Perera *et al.*, 2019). Two recently published systematic reviews deal with the broader scope of behavioral operations and supply chain management with minimum focus on behavioral ordering and inventory decisions despite being identified as a key subtopic for future works (Fahimnia *et al.*, 2019; Schorsch *et al.*, 2017). More systematic reviews have been encouraged

to catalyze future research and innovation in supply chain management (Durach *et al.*, 2017). Bibliometric and network analysis is an extension to systematic literature reviews that aid in illustrating the status quo of the literature (Persson *et al.*, 2009). This manuscript builds on the extant systematic and bibliometric reviews that broadly focus on BOM to dive deeper into behavioral ordering and inventory decisions.

This paper systematically reviews the literature of inventory and ordering decisions that use behavioral experiments either as the primary research method or as a compliment to mathematical models that explain human behavior. We target experimental works focused on both single-period and multi-period decision contexts. A meticulous data collection and review process is employed as detailed in the following sections to ensure that all relevant publications in this domain are captured in our review. Section 2 of this paper details the systematic review methodology employed. Section 3 presents the resulting bibliometric analysis from the literature pool. This provides insights into the research that drive the field. Section 4 elicits details of the experiments conducted within the scope of this paper. Under Section 5, we use a keyword analysis to identify major research clusters that have emerged within the broader field. One observation to emerging from the findings is that the newsvendor Problem and the Beer Distribution Game are utilized as the main experimental platforms in the bulk of the research in this space. As a result, we present a thorough examination of the work based on these games. The conclusions of our systematic review are presented in Section 6 covering the key findings and future research directions.

2. Review methodology

To ensure that all relevant work is captured and analyzed in the review, we employ a systematic data collection and literature mapping approach. Systematic reviews differ from traditional reviews because they follow a replicable, scientific and transparent process which minimizes selection bias through an exhaustive literature search (Petticrew and Roberts, 2006; Tranfield *et al.*, 2003). The presence of a structured methodology that encapsulates the process of scanning, analyzing and mapping the literature, as well as writing the inferences, is imperative for an effective review (Rowley and Slack, 2004). A successful systematic review begins with defining the keywords through an iterative process (Saunders *et al.*, 2009).

2.1 Defining the keyword structure

The keyword structure is defined to reflect the four spheres that constitute this review: (1) ordering/inventory context (2) operations and supply chain relevance (3) the behavioral or decision-making focus and (4) the use of behavioral experiments. Specific search terms are systematically identified through trial and error and scrutinizing the resulting papers to ensure that all relevant papers are captured. The final Boolean search was as follows: ((order* OR "beer distribution" OR "newsvendor" OR "inventory") AND (operation* OR "supply") AND (decision* OR behavior*) AND (experiment* OR "laboratory")). Using the "title, abstract and keywords" search fields of Scopus [1] search engine and limiting our search to documents published in English, our keyword structure yields 2,464 papers published through March 2020. The search is restricted to the following eight key subject areas: business management and accounting, computer sciences, decision sciences, engineering, economics, mathematics, social sciences and psychology. Further, as is standard practice among systematic reviews, our search only considers peer-reviewed journal publications (Davarzani *et al.*, 2016; Eksoz *et al.*, 2014; Fahimnia *et al.*, 2015; Perera *et al.*, 2019). Conference proceedings are not considered due to accessibility limitations and because many do not meet the criteria of being peer reviewed. Another reason not to consider conference proceedings is to avoid potential duplication of works which would've been the presented at conferences prior to publication at a journal (Eksoz *et al.*, 2014).

After initial trials of the search function, it became apparent that the keyword “order” captures a myriad of works that are irrelevant to the scope of this research. This is due to multiple meanings associated to the word “order” in the English language. However, given the significance of this keyword to our study, we retain it as a keyword in the first tier. This confounds our search process adding complexity to filter down to the most relevant papers. Works that deviate from the core of our study were identified through exclusion keywords. The exclusion words are incorporated into our keyword structure, and each paper in the resulting output is carefully analyzed to justify their exclusion. This process reduces the literature pool to 268 papers. The exclusion keywords are mainly in relation to areas that are out of scope of this review. A few examples include biomass, fuel cells, robotics, fluid dynamics, forestry, energy resources, electric machinery, hardware, induction motors and heating. The final output of the search attempts is exported and saved in comma-separated values (CSV) file format (containing all relevant information such as title, author names and affiliations, abstract, keywords and references) for further analysis.

2.2 Refining the search results

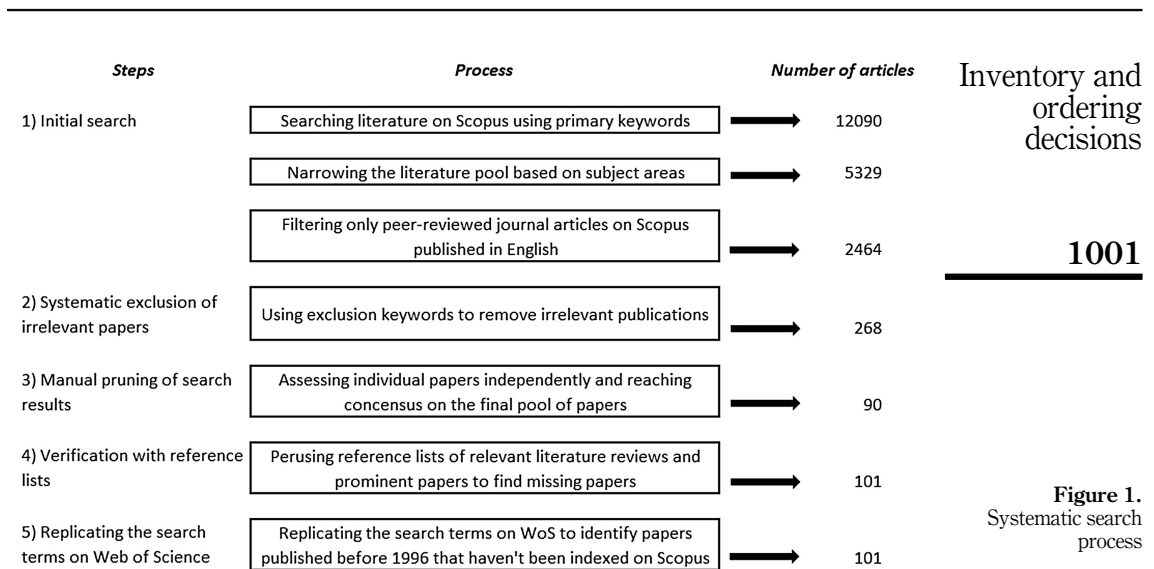
Closely inspecting the pool of 268 papers reveals that even after applying the exclusion words, there remains many papers beyond the scope of this review. As a result, we individually review titles, abstracts, and often full papers to further refine the results. This process yields 90 relevant papers. The papers that are excluded use methodologies that are outside the premise of this review or contain a normative approach. There are a handful of papers which represent different domains of BOM, such as procurement auctions, supply chain contracts, revenue management and capacity allocation.

The 90 papers deemed to be relevant through this process were cross checked with reference lists of significant preceding works in this research domain. More specifically, we cross checked the reference lists of the two review chapters mentioned earlier ([Becker-Peth and Thonemann, 2018](#); [Kremer and Minner, 2008](#)) and seminal literature ([Benzion et al., 2008](#); [Bolton and Katok, 2008](#); [Bolton et al., 2012](#); [Bostian et al., 2008](#); [Croson and Donohue, 2006](#); [Lurie and Swaminathan, 2009](#); [Schweitzer and Cachon, 2000](#); [Steckel et al., 2004](#)) and publications over the past couple of years ([Castañeda and Gonçalves, 2018](#); [D'Urso et al., 2017](#); [Schultz et al., 2018](#); [Stangl and Thonemann, 2017](#); [Tokar et al., 2016](#); [Villa and Castañeda, 2018](#); [Zhang and Siemsen, 2019](#); [Zhao and Zhao, 2018](#)). This uncovers 11 papers that are not captured by our original search terms. These papers are manually added to the literature pool, bringing the total to 101. Examination of these works reveal that they were not identified during the search process due to inconsistent keyword usage, or not listing any keywords whatsoever. One of these papers is not indexed on Scopus. To further ensure against missing work that is relevant to our review, the same keyword structure is used to replicate a search attempt in the Web-of-Science (WoS) database. No additional papers were identified.

To aid in the reproducibility of our results among interested researchers, we provide an illustration of the procedures that were followed in our literature search, along with article counts at each step, in [Figure 1](#).

2.3 Limitations of the search

The search focuses solely on eight subject areas (refer 2.1). While these eight areas represent those most closely related to operations management (OM), and thus most likely to contain relevant papers, it is possible that some work that should have been included escaped our notice. Relevant work might also have been missed by limiting our search to work published in English. It would be challenging if we were to consider other languages in the systematic review. It would be practically difficult to capture relevant content from title, abstract and



keyword given that translations were required from other languages. Further, inconsistent use of keywords in published articles creates a confound in capturing all the relevant papers using only the search and exclusion terms. In our review, this prompted a measure of human intervention to ensure that the paper pool is indeed complete, opening the possibility of some degree of individual bias influencing the results. This refers to the work of the three co-authors assessing all articles to include in the paper pool.

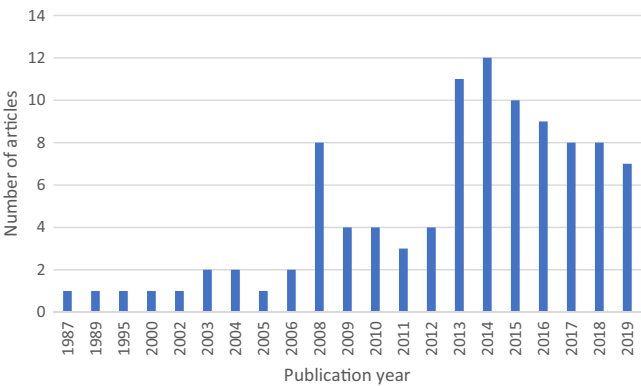
3. Bibliometric analysis

To conduct a bibliometric analysis of our literature search results, we utilize VOSviewer [2], which is a highly flexible and robust software package designed explicitly for this purpose. It allows for various methodical analysis that can be visually represented, which facilitates examination of results. We use the CSV file generated from the final literature pool as the input to VOSviewer. The CSV file contains the data fields that capture authors, affiliations, paper title, journal, keywords, citations and year of publication, which are all analyzed and presented under this section. The details pertaining to the methodologies (behavioral experiments), presented at the end of this section, were extracted manually from the papers.

We begin by looking at the growth of literature in the field over time, illustrated in Figure 2. Results show that the field has seen notable growth in volume of work published beginning in the mid noughties.

Major spikes in the number of papers in a given year as compared to immediately preceding years, such as in 2008 and 2013, are likely explained by the publication of a special issue on BOM by a major journal (2008 – *Manufacturing and Service Operations Management*, 2013 – *Journal of Operations Management*, and *International Journal of Production Economics*). Intriguingly, in the years following these special issues, the mean number of published lab experiments on ordering decisions appears to shift upwards. This could indicate that these special issues are important in generating interest amongst researchers in conducting such work. The past three years have recorded what appears to be a slight drop in the number of publications from the peak in 2014. It is too early to determine if this represents a reversion to late-2000s rates of output or if this is simply natural variance around a mean established post-2013. Taken as a whole, however, Figure 2 clearly indicates

Figure 2.
Trend in publications
over time



the growing popularity of this topic within the OR/MS community (Bendoly *et al.*, 2006; Croson *et al.*, 2013; Donohue and Siemsen, 2010; Hopp, 2004).

The next characteristic of our literature pool that we examined was the publications in which articles appeared. We find that *Production and Operations Management* and *Management Science* are the most prominent journal outlets for this topic of research, together publishing 40% of all articles on behavioral and ordering decisions (23.8% and 15.8%, respectively). 75% of all published articles can be accounted for by including six other journals: *International Journal of Production Economics*, *Manufacturing and Service Operations Management*, *Journal of Operations Management*, *Journal of the Operational Research Society*, *Decision Sciences* and *Journal of Purchasing and Supply Management*. What is interesting is that a range of journals covering the fields of operations research (OR), OM and management science (MS) have welcomed works in this area and that this openness has persisted over time.

4. Behavioral experiment methods

In this section of our review, we shift focus from the publication counts and journal outlets to an examination of the experimental methods utilized in the research we identified. After examining all 101 papers, we find that 95 of them are based solely on controlled laboratory experiments. Figure 3 illustrates the details of our results.

Beyond the lab experiments, two papers each are based on verbal protocol analysis (Cui *et al.*, 2013; Gavirneni and Isen, 2010) and vignette-based experiments (Snijders *et al.*, 2003; Tazelaar and Snijders, 2004). One paper employed a field experiment (Craig *et al.*, 2016). Finally, one paper utilized both laboratory experiments and neuroimaging studies (Zhao *et al.*, 2016). In the following subsections, we elaborate on each of these methodologies. Table A1 provides details concerning the method employed by each of the studies in our literature pool.

4.1 Laboratory experiments

Laboratory experiments are an empirical method of research as they involve observing behavior and analyzing the resulting data (Kagel and Roth, 1995; Siemsen, 2011; Smith, 1976; Tokar, 2010). They complement other methodologies by connecting analytical models with real-world problems (Deck and Smith, 2013; Katok, 2018). Laboratory experiments are ideal for discerning the ways in which human decision-makers are likely to behave in the real-world under specific conditions (Deck and Smith, 2013; Katok, 2011). Deck and Smith (2013) emphasize that well-designed laboratory experiments are highly rigorous and worthy of

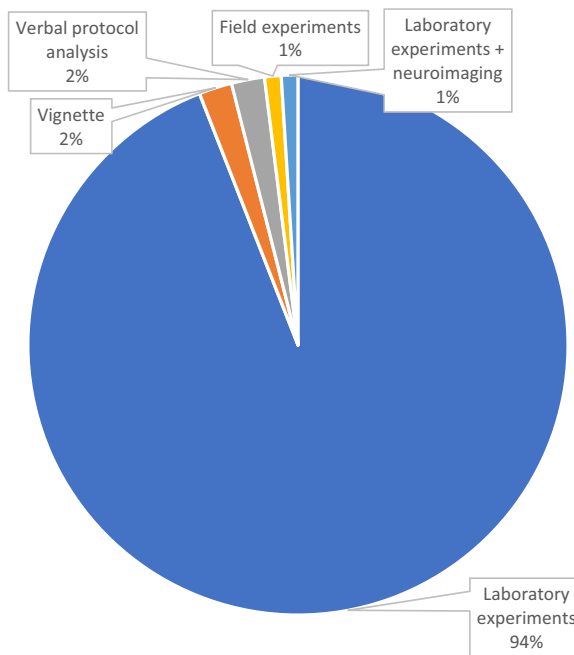


Figure 3.
Breakdown of
experimental methods
deployed in the
literature pool

utilization despite common scrutiny regarding realism. In fact, many suggest that laboratory experiments are the best methodology for establishing causality in an operational setting (Katok, 2018; Sekaran, 1992), a characteristic essential for testing and building theory.

Siemens (2011) states that laboratory experiments are an excellent option for research due to their efficiency in terms of costs and time. This is particularly valuable for groups such as doctoral students, who are often on tight deadlines attempting to complete their dissertations on schedule. More importantly though, the low cost of making changes affords researchers the opportunity to easily update or modify designs as they learn from current experiments (Katok, 2011; Siemens, 2011), allowing them to abandon unsuccessful manipulations, or find interesting discoveries.

A rich body of literature has been developed in the OM space that provides insights on how to design and conduct a laboratory experiment (Bachrach and Bendoly, 2011; Deck and Smith, 2013; Katok, 2011, 2018; Siemens, 2011). Interested researchers are encouraged to make use of these resources.

4.2 Field experiments

Field experiments are an empirical research method that can be used to explain decision effects in the real-world (natural environment), as opposed to laboratory experiments (Harrison and List, 2004; List *et al.*, 2011). Field experiments attempt to establish a cause and effect relationship within the natural environment where employees operate (Sekaran, 1992). Despite this, field experiments should not be considered as a substitute for laboratory experiments, as it is a unique methodology (Deck and Smith, 2013). Similarly, one should not confuse field experiments with field studies or case studies, which refer to correlation studies done at organizations.

Field experiments are generally costlier and more time consuming in comparison to laboratory experiments. Researchers have to be careful not to flounder the data collection as redoing the experiment would be very difficult or even impossible (Ibanez and Staats, 2018). The literature posits that there should be a justifiable theoretical motivation to run a field experiment (Card *et al.*, 2011; List, 2011). There should also be an understanding on how a field experiment better answers the research question as opposed to other methodologies (Al-Ubaydli and List, 2015). We invite those who might be interested on learning how to design and conduct field experiments to refer relevant literature (Al-Ubaydli and List, 2015; Harrison and List, 2004; Ibanez and Staats, 2018; List, 2011; List *et al.*, 2011).

4.3 Verbal protocol analysis

Verbal protocol analysis is a more subjective and richer methodology to discern the process of human judgment and decision-making. This method requires subjects to “think aloud” while participating in their experimental task. Their remarks are audio recorded and analyzed to discern patterns in decision-making. It is practically difficult for the subjects to articulate their thoughts simultaneously while endeavoring to participate in a task. It is also possible that articulating their thoughts itself may alter their thought process (Croson *et al.*, 2013). Literature shows that such criticism can be countered by implementing correct protocols (Ericsson and Simon, 1993; Payne, 1994). Thus, this method requires the researchers to be experts in it to ensure that the decision makers’ thought process is accurately captured and analyzed. Two papers in the literature pool for this review is based on verbal protocol analysis (Cui *et al.*, 2013; Gavirneni and Isen, 2010).

4.4 Vignette-based research

Vignette-based experiments generally involve a researcher presenting a contextual scenario to subjects, typically via written text, and then asking for responses to a series of questions (Dhaigude *et al.*, 2019; Rungtusanatham *et al.*, 2011). These responses usually involve making a choice among presented options or rating a statement on a Likert scale (Croson *et al.*, 2013). This method is commonly used in consumer research, and there are examples in behavioral operations and supply chain management research (see, for example, Cantor *et al.*, 2014; Chen *et al.*, 2016; Hora and Klassen, 2013), but only two papers in our literature pool utilizes this method Snijders *et al.* (2003) and Tazelaar and Snijders (2004). This could possibly indicate an opportunity for researchers interested in ordering behavior to make use of an underutilized method, but it might also reflect some important limitations regarding vignette experiments. For example, when studying inventory and ordering decisions, researchers are commonly interested in behavior over time (e.g. Schweitzer and Cachon, 2000), or the impact of factors such as training (e.g. Tokar *et al.*, 2012), or experience (e.g. Bolton and Katok, 2008). One might be able to address factors such as these to some extent through careful recruitment and selection of subjects (Croson *et al.*, 2013), but to reliably manipulate such factors between subjects through variations of a written scenario could be exceptionally difficult. Moreover, vignettes are only able to capture stated or intended behavior, not actual behavior, which can be subject to issues such as social desirability bias. In short, while vignettes are a valuable item in the toolbox of an experimental researcher, their appropriateness for research on order decisions might be limited.

5. Primary research clusters

Bibliometric studies focus on networks of keyword co-occurrences. Networks help clustering a group of well-connected papers that is less connected with the rest of the literature within the domain of study (Callon *et al.*, 1983; Davarzani *et al.*, 2016; Fahimnia *et al.*, 2015; Perera

et al., 2019; Peters and Van Raan, 1993). Keyword co-occurrences are graphically represented using network diagrams. Nodes represent a keyword while the links represent the connection between two keywords. Co-occurrences represent the frequency of times a keyword pairing has been used within a given literature pool. The higher the frequency, the stronger the link-weight that represents the connection between a keyword pairing (Leydesdorff, 2011; Van Eck and Waltman, 2010, 2014).

We extracted the keywords of our paper pool using the CSV files downloaded from Scopus. The keywords were analyzed using the VOSviewer software. We amalgamated similar keywords (e.g. behavioral operations and behavioral operations management; supply chain and supply chain management; beer game and beer distribution game), to make the analysis more pragmatic. Keywords referring to methodological aspects (e.g. experiment, experiments, behavioral experiment, laboratory experiment, verbal protocol analysis), were also excluded from the analysis. Given their frequent use, these terms confounded the clustering process.

There were 449 author and indexed keywords used across the 101 publications considered in our literature pool. Further analysis indicated that 362 of those were only used by one paper and provided scant insights in clustering the publications. Therefore, these single-use keywords were removed from the analysis, leaving 87 keywords used in two or more papers in the literature pool. After analyzing these keywords using network science principles on VOSviewer (Van Eck and Waltman, 2010), results show that some of the words used only twice (e.g. mental accounting, feedback, retailing), could not be distinctly ascribed to a sub stream of study within behavioral ordering decisions. Hence, these keywords were also removed, leaving only 43 keywords used in a minimum of three papers.

The clustering of keywords illustrates the emergence of six clusters within our literature pool, which is presented in Figure 4. Each cluster is assigned a separate color – green, red, blue, olive, purple and aqua. Further inspection of the clusters reveals that the three smaller clusters (olive, purple and aqua) can be consolidated into one given the similarity in their works and the relatively smaller size. These clusters are not mutually exclusively as papers tend to overlap different domains.

Three prominent clusters clearly emerge from Figure 4. The green cluster (Cluster 1), which is the largest, contains keywords “newsvendor problem,” “behavioral operations management” and “experimental economics.” The second largest cluster, shown in red (Cluster 2), contains the keywords, “beer distribution game” and “bullwhip effect.” The third largest cluster, shown in blue (Cluster 3), is made up of more general terms, such as “inventory management,” “decision-making” and “supply chain management.” The analysis seems to indicate the existence of three distinct clusters of research within the scope of inventory ordering decisions.

Cluster 1 contains research on the newsvendor problem. Based on the other keywords in the cluster, it seems these studies are typically being generated by BOM researchers and take an experimental economics approach (lab experiments conducted by experimental economists tend to strictly follow a set of guidelines perhaps best laid out by Smith (1982)). Cluster 2 is work focused on studies conducted primarily using the beer distribution game as an experimental platform to discern BWE. Cluster 3 (blue) is work focused on inventory decisions beyond newsvendor or beer game contexts that deals primarily with multi-period ordering decisions. The prominence of the term “supply chain management” in this cluster also speaks to the more general nature of this grouping. The fourth cluster, a consolidation of the smaller olive, purple and aqua clusters (Cluster 4), can be summarized as referring to behavioral constructs of ordering decisions. This cluster is composed of works that attempt to identify behavior as opposed to discerning the effects of human behavior on operations (e.g. Chen *et al.*, 2012; Strohhecker and Größler, 2013; Lee and Siemsen, 2017). For instance, Chen *et al.* (2012) conduct an experiment and use that data to model bounded

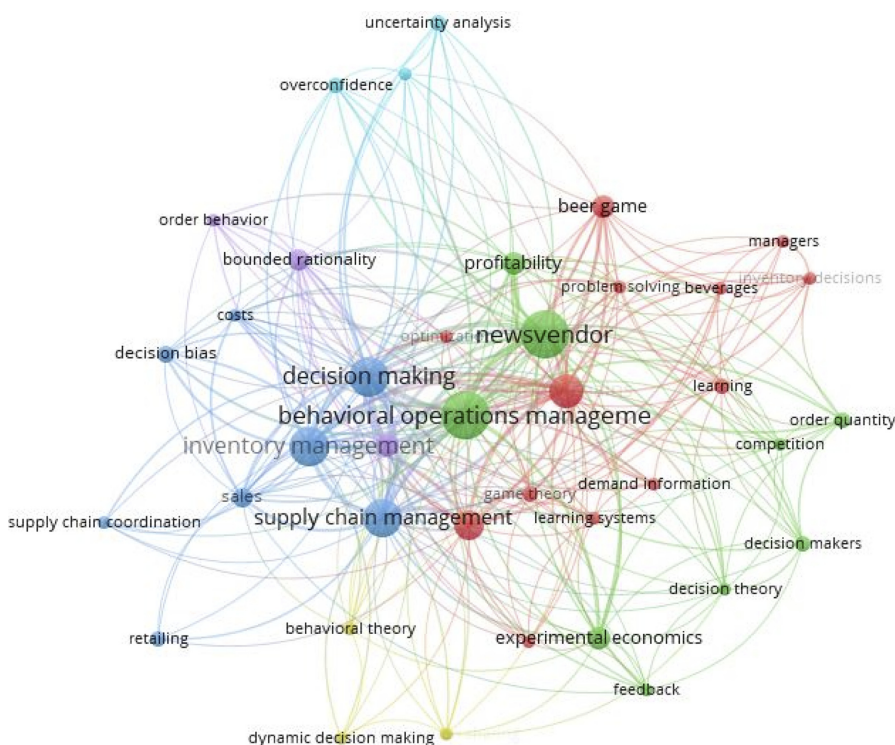


Figure 4.
The research clusters
(color coded in green,
red, blue and olive)

rationality based on the quantal response equilibrium. [Strohhecker and Größler \(2013\)](#)’s paper investigates the effect of four personal traits (intelligence, knowledge, personality and interest) on ordering decisions. [Table A1](#) provides a table that identifies the papers allocated under each cluster.

By examining the use of the keywords over time (longitudinally), we observe the evolution of these clusters. The analysis is based on the number of publications per cluster per annum. [Figure 5](#) visually represents the outcome of the analysis.

The cluster containing “beer distribution game” and “bullwhip effect” appears to be older than the others emerging primarily in the mid-noughties. Keywords seem to have shifted more towards “supply chain management” and “decision-making” shortly after 2010 but shifted again to the more specific newsvendor context later in the decade. Interested researchers could investigate this observation in greater detail, shedding light on drivers of these observed changes. It could represent a change in interest among researchers (i.e. a general preference for the specific newsvendor problem over others), but it is possible that there are more pragmatic drivers. For instance, the beer distribution game can be a noisy environment due to its design (e.g. multiple decision makers, long order lead times) sometimes limiting the ability to observe the impact of certain factors of interest despite being exceptionally useful as an experimental platform.

Some of the most recent keyword observations are part of the fourth cluster, and include terms such as “supply chain coordination,” “overconfidence” and “uncertainty analysis”. These emerging keywords represent the beginning of another shift in research focus. It is also noted that it represents a second wave of newsvendor-based research given that all of these

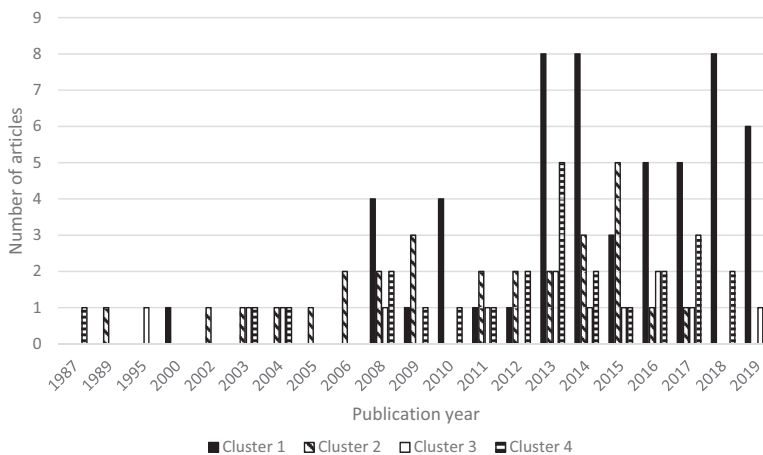


Figure 5.
The evolution of
research clusters

are issues that have been explored in a newsvendor context, but from a different angle more focused towards understanding behavioral constructs. As a result, newsvendor-based research has enjoyed a consistent focus from researchers.

5.1 Newsvendor problem

The newsvendor problem is a classic inventory problem introduced in 1951 (Arrow *et al.*, 1951). In the newsvendor problem, decision makers place orders for products that cannot be replenished during the selling season (period) and becomes obsolete at the end of the season. Its name is derived from the conundrum faced by vendors selling newspapers on how much to order. Stocks must be ordered during the previous period. Excess stocks at the end of the day become obsolete at a cost. Shortages results in a loss of possible sales (revenue). Thus, the problem hinges on balancing the cost of excess inventory with the cost of shortages (Kirshner and Ovchinnikov, 2018).

We find 56 papers under this cluster, (Appendix 1) most of which are published in *Production and Operations Management* (16), *Management Science* (10) and *Journal of the Operations Research Society* (6). Ulrich Thonemann is the most prolific author in this cluster with 5 contributions. Eight authors have made 3 contributions each in this cluster (Michael Becker-Peth, Jaime Castaneda, Mirko Kremer, Brent Moritz, Enno Siemsen, Jordan Tong, Luk Van Wassenhove and Xiaobo Zhao). Like in any field, strong and ongoing collaborations seems to be a prominent feature in putting these authors at the top. We also note a large number of authors affiliated to or connected to Germany among the leaders with the University of Cologne being a prime origin of research in this cluster.

5.1.1 Utility functions. Literature has perceived the newsvendor problem from a behavioral economics perspective where the utility function plays a dominant role. The utility function of the decision maker is dependent upon his/her objectives that expand beyond profit maximization. Thus, this is a very important avenue of research that hinges on core precepts of behavioral economics such as risk preference (risk taking or risk averse behavior), loss preference (losses carrying a heavier mental value than gains), prospect theory, mental accounting, inventory error and impulse balance (Becker-Peth and Thonemann, 2018).

Risk preferences seem to be of little effect in the newsvendor problem. Evidence shows that risk averse subjects are willing to pay to obtain additional information that aids their

ordering decisions (Kremer and Van Wassenhove, 2014), although there is scant impact on their ordering behavior (de Véricourt *et al.*, 2013; Schweitzer and Cachon, 2000). Becker-Peth *et al.* (2018) report correlation between individual risk preferences and order quantities, providing evidence that risk preferences affect order behavior. They only find a marginal moderation effect on how information about the effect of order quantities on the profit distribution affects ordering. An earlier study shows how inventory decisions are affected by incorporating reference dependent valuation into decision models (Becker-Peth and Thonemann, 2016). The findings from the laboratory indicate a deviation from normative solutions.

Prospect theory posits that humans evaluate decisions against changes from a certain reference point rather than comparing the alternatives with one another, and as a result, are prone to risk preference reversal based on the way in which the decision is framed (Kahneman and Tversky, 1979). Decision makers are typically risk averse when decisions are framed as gains, but because of humans' strong aversion to losses, they become risk seeking when decisions are framed as losses. While numerous normative works have investigated the interrelation between the newsvendor problem and the prospect theory (Long and Nasiry, 2015; Nagarajan and Shechter, 2014; Uppari and Hasija, 2017), there is little experimental evidence in this space. Schultz *et al.* (2018) is the lone example. They frame newsvendor decisions as losses vs. profits but, contrary to theory, find no significant differences in performance. We call for future works in this domain to further enrich the literature.

Mental accounting refers to the process of framing, combining and evaluating economic outcomes (Thaler, 1985). One paper investigates whether the time of payment affects newsvendor ordering decisions (Chen *et al.*, 2013). They test two payment schemes, supplier financing and customer financing. Customer financing defers from the traditional setting (supplier financing) by ensuring the customer immediately bears the cost of the units sold. Results show that the average order quantities in the customer financing treatment were significantly lower. Another study reveals that the type of payment carries weight in decision-making (Schiffels *et al.*, 2011). The study finds that opportunity costs are underweighted when compared to a scenario with a penalty on underage costs. Further evidence shows that decision makers maintain different accounts for sales and excess inventory (Becker-Peth *et al.*, 2013) and different contract types (Katok and Wu, 2009).

Ex post inventory errors have been reported to press a disutility on decision makers, prompting them to behave differently (Schweitzer and Cachon, 2000). Several works investigate the relationship of inventory errors in a newsvendor setting (Bostian *et al.*, 2008; Ho *et al.*, 2010; Kremer *et al.*, 2010). The impulse balance theory assumes that order quantities that vary from the optimal trigger impulses are proportionate to the costs resulting from the variation of the order quantity. A couple of works investigate the connection of this phenomenon on the newsvendor problem (Ockenfels and Selten, 2014, 2015).

5.1.2 Critical ratio research. The critical ratio corresponds to the profit margin applicable in a newsvendor scenario, where c_u is underage cost (demand that cannot be satisfied using inventory) and c_o is overage cost (excess inventory that cannot be sold). How human decisions vary depending on different critical ratios have been extensively researched using laboratory experiments (Bolton and Katok, 2008; Bolton *et al.*, 2012; Schweitzer and Cachon, 2000).

$$\text{Critical ratio} = \frac{c_u}{c_u + c_o}$$

Human decision-making under various critical ratios has been studied extensively (Bolton and Katok, 2008; Bolton *et al.*, 2012; Schweitzer and Cachon, 2000). Literature widely confirms a bias known as the pull-to-center effect. This describes the tendency for decision makers to order less (more) than the nominal quantity when critical ratio is relatively high (low) (Bostian

et al., 2008; Schweitzer and Cachon, 2000). This effect has been observed among both student subjects and industry practitioners, suggesting that experience has little or no influence on bias (Bolton and Katok, 2008).

In an attempt to explain the pull-to-center effect, an anchoring heuristic has often been cited (Becker-Peth and Thonemann, 2018). Literature has presented calculations of anchor factors (the extent to which the order quantity is anchored) to substantiate this claim, finding that subjects tend to anchor on mean demand and insufficiently adjust toward the optimal order quantity (Bolton *et al.*, 2012; Schweitzer and Cachon, 2000). Anchor factors have been used to differentiate behavior based on considerations such as cognitive reflection (Li *et al.*, 2019; Moritz *et al.*, 2013) and subject experience (Bolton *et al.*, 2012). Su (2008) proves that quantal choice models (order quantities are chosen randomly) can explain the pull-to-center effect when the demand distribution is bounded. However, other literature suggests that the quantal choice model cannot fully explain the pull-to-center effect because it exists under normally distributed demand, which isn't bounded (Benzion *et al.*, 2008). Recent research proves that the pull-to-center effect exists with strategic customers too. Research also reveals that well-designed radical system recommendations provided by decision support systems may eliminate the pull-to-center bias (Feng and Gao, 2020). Additionally literature posits that when the unit cost is low/high, the pull-to-center bias reduces/increases (Zhang *et al.*, 2018). They call for future research to consider pricing as a decision variable in addition to inventory.

A recent study that analyses 24 newsvendor experiments finds that design factors of experiments drive the pull-to-center asymmetry (Zhang and Siemsen, 2019). They argue that if the pull-to-center effect is purely a result of human behavior, the asymmetry observed in different studies should be consistent. They posit (1) how likely it is to make a loss in high or low margin conditions, (2) how salient the underage costs are framed, and (3) whether a decision support system exists are three factors determining the presence of the pull-to-center effect. The research concludes that the pull-to-center effect is an empirical phenomenon while the asymmetry of it is not (Zhang and Siemsen, 2019). Interestingly, the authors recommend sharing information on underage loss is advantageous in a high margin setting while recommending not to do so in low margin settings.

To mitigate this decision biases (and potentially others), literature suggests three possible approaches; (1) learning, (2) training and (3) decision support (Becker-Peth and Thonemann, 2018). Bostian *et al.* (2008) observe how learning influences the pull-to-center effect by calculating the anchor factors at the beginning and the end of an experiment with 30 ordering decisions. They report that only a few previous periods and optional order quantities are considered by a boundedly rational decision maker when determining the order quantity. This phenomenon is posited as a plausible reason for the pull-to-center effect. Other works examining a sequence of newsvendor decisions did not find evidence of learning (Bolton and Katok, 2008; Ockenfels and Selten, 2014).

Bolton *et al.* (2012) investigate whether training may mitigate the pull-to-center effect. Their results show that it does improve decision-making relative to no training, but that orders are still significantly different from the optimal, meaning that it did not eliminate the bias. Some degree of decision support is examined in nearly all newsvendor experiments. Bolton and Katok (2008) provide subjects with additional information on expected profits and foregone profits in their experiment. They also try limiting subjects' options regarding order quantities. Neither results in a significant improvement in performance. Bolton *et al.* (2012) went further by expressing the nominal order quantity and the expected profit curve as a function of the order quantity. Although subjects' order quantities moved towards the nominal amount, there was still a significant difference between them. More work on the role of the learning effect to mitigate the pull-to-center effect is encouraged.

Additionally, we note that some research questions whether or not the pull-to-center effect even truly exists or is simply an artifact of data aggregation. [Lau et al. \(2014\)](#) study the newsvendor model on an individual level (as opposed to averaged group-level), and find that decision-making is highly heterogeneous, and that predictions of behavior based on the pull-to-center effect are largely inaccurate.

Behavioral research on the newsvendor problem has often relied on students. We encourage researchers to experiment outside of classrooms in the real-world for a wider impact. One such research on the utilization of operating theatres in German hospitals uses medical professionals as subjects hinge on the fundamentals of the newsvendor model ([Fügener et al., 2017](#)). Similar applications of the newsvendor model into real-world problems will be fertile ground for future research.

Extant literature has relied heavily on uniform and normal distributions. We encourage the use of other demand distributions to align more closely with industry problems. [Fügener et al. \(2017\)](#) use lognormal distributions in addition to uniform distributions to relate to utilizing operating theatres. It is equally important to venture into researching the pull-to-center asymmetry through decision error (percentage error) as opposed to comparing with the optimal order quantity ([Zhang and Siemsen, 2019](#)).

We encourage further investigation on decision support in a newsvendor setting. Effects of uncertain events such as supply chain disruptions, retail promotions and environmental influences can be coded into behavioral experiments. Such works would provide insights into how real-world supply chains can better utilize the information to support decision-making in a newsvendor setting.

5.1.3 Other heuristics and drivers of newsvendor behavior. Heuristics are a practical approach used to solve complex problems. While the error they produce is generally tolerable (hence their usage), it is often systematic, thus leading to biased decision-making. As previously discussed, the pull-to-center effect is often suggested to result from the use of an anchoring and adjustment heuristic, where the adjustments tend to be insufficient to approach optimal and are consistently biased given a certain set of decision parameters.

The literature suggests other uses of the anchoring and adjustment heuristic and resulting biases. The demand chasing heuristic is one such example. This occurs when decision makers anchor on previous order quantities and adjust them towards previous demand realizations ([Becker-Peth and Thonemann, 2018](#)). The common observation is that low-demand realizations lead to lowering order quantities and vice versa ([Bolton et al., 2012](#)). Research shows that Chinese subjects are more prone to chase demand compared to their American counterparts. This results in a lower expected profit and greater variance of ordering quantities ([Li et al., 2019](#)). Demand chasing can be reduced when the subjects have to place orders for multiple periods as opposed to a single period ([Bolton and Katok, 2008](#)).

Four different calculation methods have been deployed in the literature to report the demand chasing heuristic: (1) change frequency ([Benzion et al., 2008](#); [Schultz et al., 2018](#); [Schweitzer and Cachon, 2000](#)), (2) adjustment score ([Schweitzer and Cachon, 2000](#)), (3) regression ([Bostian et al., 2008](#); [Moritz et al., 2013](#)) and (4) correlation ([Bolton and Katok, 2008](#); [Lee and Siemsen, 2017](#)). These four methods are mathematically evaluated for accuracy ([Lau and Bearden, 2013](#)). The findings recommend using correlation to report demand chasing in future works as they are less prone to falsely detecting demand chasing behavior.

In sum, the newsvendor problem has provided a platform for a rich body of research in the inventory and ordering behavior space, and it shows promise to continue offering research opportunities that shed new light on human behavior. As researchers have formulated new and interesting questions since the seminal work of [Schweitzer and Cachon \(2000\)](#), the basic newsvendor problem has been expanded to incorporate additional elements of replenishment decisions. One example is the multilocation newsvendor problem, in which the decision

maker is required to make multiple/independent newsvendor decisions (Ho *et al.*, 2010). In addition, a newsvendor problem in which the decision maker must contend with multiple products has also come to the fore (Castañeda and Gonçalves, 2018). Among the most prominent of these new developments is the competing newsvendor problem. As a result, we review the literature regarding this context in the subsequent subsection. These new paradigms are fertile ground for future research.

5.1.4 Competing newsvendor. The competing newsvendor problem is an advanced version of the traditional newsvendor problem, which is monopolistic in nature. This variant of the newsvendor problem instead considers a competitive environment, where multiple newsvendors compete with one another to satisfy consumer demand (Kirshner and Ovchinnikov, 2018). This brings into play human-to-human or human-to-computer interaction to maximize individual profits (Feng and Zhang, 2017). Evidence suggests that humans behave differently in a competitive environment as opposed to a monopolistic environment (Kirshner and Ovchinnikov, 2018).

Research utilizing the competing newsvendor problem has only recently appeared in the literature (Ovchinnikov *et al.*, 2015). Since then, this subdomain has received considerable attention (Feng and Zhang, 2017; Quiroga *et al.*, 2019; Zhao and Zhao, 2016; Zhao and Zhao, 2018). All these studies are focused on a duopolistic market (Kirshner and Ovchinnikov, 2018). The first experimental work in this space saw subjects competing with an automated newsvendor in two duopolistic markets at the same time. The experiment was conducted for independent and identical demand (iid). The findings show that the human subjects failed to sense the competitor's strategy and respond accordingly (Ovchinnikov *et al.*, 2015).

Experiments that pitted human subjects against one another as competitive newsvendors soon followed suit (Feng and Zhang, 2017; Zhao and Zhao, 2016). Feng and Zhang (2017) find that subjects in a duopolistic environment tend to accumulate more inventory than subjects in a monopolistic environment. This was observed for a high margin scenario in a previous research as well. Low margin scenarios reported an increase in order oscillations (Zhao and Zhao, 2016). Their findings indicate that subjects tend to ignore competitors and give minimal consideration on the overflow demand.

The pull-to-center effect has also been observed in competitive settings (Feng and Zhang, 2017; Ovchinnikov *et al.*, 2015; Zhao and Zhao, 2016). Under incomplete information, this effect was only observed in a low margin setting (Zhao and Zhao, 2018). Findings from behavioral experiments on the competitive newsvendor problem have also been extended to build behavioral models to elucidate human decision-making (Zhao and Zhao, 2016, 2018). Another study evaluated profit differences between behavioral and management science-driven competitors. The results show human decision-makers obtaining a substantially lower profit than the equilibrium expected profit (Quiroga *et al.*, 2019).

Given the relative novelty of this special case of the newsvendor problem, there is a lot more that could be experimentally investigated. There's fertile ground for research beyond duopolistic competition. Despite the real-world supply chains being competitive, majority of the extant experimental literature has failed to capture that. Running experiments on various facets of the competing newsvendor problem is likely to provide insights on human behavior in a competitive environment. Understanding how decision makers behave under a competitive environment for the newsvendor problem is likely to shape the research agenda for competitive multi-period ordering decisions in the future.

5.2 Beer distribution game

BDG was first introduced at the Massachusetts Institute of Technology (MIT) in the late 1950s by Forrester (1958). Playable on a board or computer, it has grown over the decades to become a popular tool for illustrating some of the fundamental challenges associated with

supply chain management, particularly BWE. It is commonly used in both the classroom and executive education settings. Traditionally, the game includes players who represent four echelons in a hypothetical beer supply chain (production plant, distributor, wholesaler, and retailer). Gameplay involves the players collaborating in an attempt to fulfil customer demand while minimizing costs (Croson *et al.*, 2014; Sterman, 1989).

Sterman (1989) is the first to conduct academic research using an experimental BDG. It is widely considered a seminal piece of research in the field of BOM, which subsequently adopted the BDG for additional studies on inventory and ordering decisions as it emerged in the noughties (Croson *et al.*, 2014; Croson and Donohue, 2002, 2006). The traditional four echelon BDG has been most commonly employed (Ancarani *et al.*, 2013; Croson *et al.*, 2014; Croson and Donohue, 2005, 2006; Hung and Ryu, 2008; Niranjana *et al.*, 2011; Ouyang, 2014; Wu and Katok, 2006). However, scaled-down versions using only three echelons (Steckel *et al.*, 2004) or two echelons (Cantor and Katok, 2012; Özer *et al.*, 2014; Wang and Wang, 2010; Wu, 2013) have also been used. We also observe literature on supply chain contracting that sometimes use a two-echelon lab experiment (Davis *et al.*, 2014; Davis and Hyndman, 2019). The number of echelons can be easily altered to suffice the research questions. Several publications have relied on a combination of BDG and other multi-period inventory experiments (Tokar *et al.*, 2011, 2012, 2016).

Exploring BWE has been the prime driver behind the body of BDG literature (Ancarani *et al.*, 2013; Croson and Donohue, 2006; Niranjana *et al.*, 2009; Rong *et al.*, 2008). Exploring causes of the BWE, or strategies for mitigating the BWE, has been the chief objective of this cluster. Experiments can rely on having human subjects in each echelon of the supply chain or automating some of the echelons, depending on the expectations of the research (Ouyang, 2014). Analyzing the ratio between the order variances of each adjacent echelon is commonly deployed to ascertain the BWE (Croson and Donohue, 2006).

The nexus of Rachel Croson and Karen Donohue has contributed prolifically with 5 papers out of the 28 papers in the BDG cluster (Appendix 1). John Sterman and Elena Katok have contributed 3 papers each to the cluster. The most significant outlet for this cluster is *International Journal of Production Economics* (5) followed by *Management Science* (4) and *Production and Operations Management* (4).

5.2.1 Causes of the BWE. The seminal paper by Lee *et al.* (1997) lists four key causes for the BWE: (1) demand forecasting, (2) rationing and shortage gaming, (3) order batching and (4) price fluctuations. It is important to highlight that within this domain there are structural and the behavioral causes that affect the BWE. In the practice there is a dearth of information which leads to suboptimal decisions (Quiroga *et al.*, 2019). Academics must acknowledge that even rational decision makers will cause the BWE (due to variations observed in the above four causes beyond their control). It is also noted that humans tend to increase the structural biases in addition to the behavioral biases they possess.

In Sterman's aforementioned seminal paper, he identifies decision maker's propensity to ignore time delays and give inadequate consideration to their supply line as a behavioral bias leading to the BWE (Sterman, 1989). This bias has come to be known as supply line underweighting (SLU) (Croson *et al.*, 2014; Croson and Donohue, 2006; Niranjana *et al.*, 2011), and is characterized by the failure of decision makers to consider orders that have already been placed but are yet to be received (pipeline inventory). This bias leads to order variations as decision makers attempt to manage their inventory levels causing the BWE.

SLU is thought to result from the cognitive limitations of decision makers (Sterman, 1989). Indeed, Narayanan and Moritz (2015) report that the likelihood of underweighting the supply line is dependent upon the cognitive reflection level of the decision makers. SLU has been studied in conjunction with other behavioral ordering issues, such as overordering and coordination risk. Research suggests that overordering leads to SLU (Niranjana *et al.*, 2011), but shows that coordination stock is ineffective (Croson *et al.*, 2014).

Another behavioral bias causing the BWE is coordination risk. Coordination risk refers to the uncertainty managers' face regarding the behavior of their supply chain partners, leading to uncertainty regarding the collective outcome (Croson *et al.*, 2014). This uncertainty leads to variation in ordering behavior, which ultimately causes BWE. Croson *et al.* (2014) investigate this phenomenon and examine multiple means of addressing BWE, including the practice of holding excess inventory on-hand, generating common knowledge, and informing supply chains of the optimal ordering policy. The same dataset has been used in a subsequent study that investigated phantom ordering. Phantom ordering refers to ordering excess to what is actually required to counter supply uncertainty (Sterman and Dogan, 2015). The study reveals that 22% of the subjects ordered 25 times more than the known, constant demand. The authors surmise that environmental stressors such as large orders, backlogs or late deliveries trigger such behavior.

5.2.2 Mitigating the BWE. Information sharing between different supply chain roles have been extensively studied as a way to mitigate BWE (Croson and Donohue, 2005, 2006; Ouyang, 2014; Sarkar and Kumar, 2015; Zhao and Zhao, 2015). Early works using BDG report that sharing information about downstream (closer to the final customer) partners' on-hand inventory amounts with upstream partners (away from the final customer) helps mitigate order oscillations across the supply chain. The same results were not observed when sharing upstream information with downstream partners (Croson and Donohue, 2005). Inventory information sharing has also resulted in upstream partners being more prepared to counter BWE (Croson and Donohue, 2006). Both these works suggest that information sharing is more likely to benefit upstream partners (Croson and Donohue, 2005, 2006). Evidence also suggests that advanced demand information sharing helps mitigate BWE (Ouyang, 2014). Zhao and Zhao (2015) conduct an experiment where they compare full information sharing with no information sharing and find no significant evidence of information mitigating BWE. They report that the subjects with access to full information attempted to reduce stockouts while those with no information were more focused on reducing inventory. Finding the balance between the right kind/amount of information is vital. More research is required to answer this debate.

The effect of sharing retail demand data, point-of-sales (POS) data, on BWE has also been investigated. This has yielded mixed results. Croson and Donohue (2003) report that POS data reduces some components of BWE in a stable demand setting while Steckel *et al.* (2004) find that it is likely to distract decision makers (especially farther from the end customer), causing them to ignore more pertinent information.

Other notable studies related to information sharing and order variation include Cantor and Macdonald (2009), who explore the role of problem-solving approaches (abstract vs concrete), in the use of information as defined by the construal level theory. Production smoothing and seasonality has also been a focal point regarding BWE mitigation. Cantor and Katok (2012) find that predictable seasonal ordering leads to smoothed retail orders. Applying a cost to changing order quantities further reinforces this, leading to a disappearance of BWE in some settings. Sarkar and Kumar (2015) explore the specific context of supply chain disruptions and find that information sharing is critical to normalizing operations. Disruptions proved to lead to increase order variability. Experimental evidence indicates that manufacturers who inform their supply chain partners about disruptions tend to reduce BWE. However, they observed that passing information on disruptions at retailers or distribution centers does not significantly reduce BWE. The reverse bullwhip effect (RBWE) refers to the opposite of BWE where order variability increases downstream. Experimental evidence and simulations find that RBWE occurs when there are upstream disruptions and supply overweighting (Rong *et al.*, 2008).

Despite being one of the most utilized experiments in BOM, BDG still has a lot to offer. There are ample research opportunities to identify how to mitigate BWE and SLU.

Understanding the effects of various operational challenges (e.g., retail promotions, supply uncertainty, capacity constraints) are fertile ground for future research. We see potential to bring a competitive dynamic to at least one echelon of the BDG to better understand decision-making under competition and how other echelons adapt to it. The supply net game provides a good platform for this as it mimics a real-world supply net as opposed to a serial supply chain (Delhoum and Scholz-Reiter, 2009). However, future researchers need to be cognizant of the complexities associated in running BDG as finding subjects in the multiples of the number of echelons in the experiment and designing an interactive experimental software could be cumbersome.

5.3 Multi-period order decision platforms

In addition to the newsvendor game and BDG, other multi-period platforms are being utilized by researchers. Like the newsvendor game, they tend to focus on individual decision makers (as opposed to teams in the BDG). Unlike the newsvendor game, these tasks often allow for inventory to carry over from one period to the next, making them reflective of a broader range of consumer products in the marketplace. Thus, making it comparatively more complex to design and implement.

The attention received by this cluster is scant with only 14 papers (Appendix 1). Three papers each have been published on *Production and Operations Management* and *Manufacturing and Service Operations Management*. The research collaboration between Travis Tokar, John Aloysius and Matthew Waller has been significant in shaping this cluster with three papers. The 14 papers in this cluster have come from 32 authors underscoring the opportunity for more researchers to engage in this cluster.

Initial works juxtapose decision-making in a single-period and multi-period settings and report significant differences (Bloomfield and Kulp, 2013). It was evident that transit lags and durability of products (multi-period nature of the product) exacerbated decision suboptimality. They report that order volatility increases in multi-period ordering settings. This is because the decision makers insufficiently adjust order quantities to reflect inventory levels and backlogs. Transit time was reported as a major source of suboptimal ordering decisions, more so than in a newsvendor setting.

Tokar *et al.* (2014) investigate bracing behavior in ordering decisions. Bracing refers to biased preparation behavior resulting from an overestimation of the likelihood of a negative event. They specifically examine ordering decisions in response to uncertainty about the timing and magnitude of a looming demand spike. Their results indicate the subjects have more difficulty efficiently managing uncertainty of timing than uncertainty of the volume.

A recent lab-based study investigated the impact of budget cycles towards multi-period inventory decisions (Becker-Peth *et al.*, 2020). The findings show a sharp drop in order-up-to levels at the conclusion of the budget cycle indicating a myopic nature among subjects. The paper reports that an increase in starting inventory when a starting inventory factor is introduced in the experiment provide the subjects a better appreciation of managing for the current and future budget cycles.

The importance of sharing information about the timing and magnitude of a possible demand spike, such as a retail promotion, in reducing relevant supply chain costs had been identified in preceding literature (Tokar *et al.*, 2011). From a supply chain perspective, creating incentives that align with the objectives of forecasting and ordering decision is paramount. Although the effect of incentives has been behaviorally investigated in a newsvendor setting (Scheele *et al.*, 2017), applying this to a multi-period ordering setting is highly pertinent. Both lack of supply to meet the demand (Villa *et al.*, 2015), and delivery delays and feedback were found to inflate order amounts. It was also found that order quantities take a saw-tooth pattern of overordering and underordering until an equilibrium is

reached. These findings demonstrate the significance of carefully planning promotions by retailers and communicating them to supply chain partners.

Literature has attempted to discern how traits, experience and education may affect inventory decisions. Bloomfield and Kulp (2013) study the need for cognition, impulsiveness and locus of control and find little evidence on their relevance on ordering decisions. Influence of four personal traits (intelligence, knowledge, personality and interests) on ordering decisions have also been studied (Strohhecker and Größler, 2013). The findings show intelligence as the strongest predictor of ordering performance. The other factors show significant but weak relationships with ordering performance. Strong interest in social issues lead to incurring excess costs. They identify that eagerness to gain new experiences also leads to excess costs.

The economic order quantity (EOQ) model is a classic inventory management approach that prescribes how much to order and when to order such that the sum of inventory holding cost and ordering cost is minimized (Erlenkotter, 1990; Harris, 1913). It is a multi-period ordering setting which has ample space for contribution in behavioral operations (Becker-Peth and Thonemann, 2018). The sole experimental work in this space investigates the effect of equivalent performance metrics on inventory decisions (Stangl and Thonemann, 2017). They report that the inventory decisions are significantly affected by the inventory metric employed. Given that the newsvendor problem has inspired such an abundance of behavioral research projects, we wish to call attention to the tremendous opportunity that exists to apply a behavioral focus to this widely studied, classic inventory model (Erlenkotter, 1990).

In sum, after examining the literature, we encourage more research on multi-period ordering decisions. Not only might this create knowledge and build theory in new areas of inventory management, but it also stands to have tremendous practical implications. Works focused on ordering behavior regarding such issues as retail promotions, supply disruptions, and supply/demand uncertainty could all be of benefit to industry practitioners. Further, companies are very mindful of maintaining high service levels to satisfy customer demand, and these high levels have been shown to alter ordering decisions (Craig *et al.*, 2016), presenting yet another important research context in this space.

5.4 Behavioral constructs of ordering decisions

Leveraging behavioral experiments, the papers in this fourth cluster dive deep into human decision-making to test or explain behavioral constructs and/or psychological theories. All of these papers are connected to the previous clusters given that the core experiment involves either a single-period/multi-period inventory decision or a BDG. Some of these works have already been discussed in the previous sections given the overlaps in focus. Thus, the most salient details that had not been discussed previously are presented under this section. We foresee this cluster growing in the future as experimental research can be used to elicit effects of demographic or psychological factors.

We record 29 papers in this cluster (Appendix 1). *Management Science* (8), *Production and Operations Management* (5) and *Journal of Operations Management* (4) are the most prolific outlets to contribute to this cluster. Ulrich Thonemann emerges as the most significant author in this cluster alongside Yefen Chen with three papers each. It is evident that this cluster has seen contribution from multiple authors and is not dominated by a few like the other three clusters. This variety is salient towards investigating the practical and academic implications of this research cluster further.

BOM research has focused heavily on heuristics and biases. Several examples of this were presented under Section 5.1. This literature contains research that is primarily focused on theories from psychology. The cognitive reflection theory (Moritz *et al.*, 2013), impulse balance theory (Ockenfels and Selten, 2014) and the construal level theory (Cantor and

MacDonald, 2009) have been three popular lenses for studying behavior. We also find several works in the literature which attempt at proving/building theory by using mathematical models complimented by behavioral experiments (Ho *et al.*, 2010; Su, 2008). This is a challenging research domain that needs to be further appreciated and encouraged. We note an increasing trend in literature that are using behavioral experiments to prove mathematical models (Scheele *et al.*, 2017; Zhang *et al.*, 2018; Zhang and Siemsen, 2019). Majority of the newsvendor problem literature in our paper pool follow this pattern in aiming to prove/provide mathematical models.

Overconfidence has been cited as a potential source of the pull-to-center effect in the newsvendor problem (Lee and Siemsen, 2017). Broadly speaking, this results from the finding that decision makers underestimate demand variance. Ren and Croson (2013) show that nearly 30% of the pull-to-center effect can be explained using overconfidence, and that methodically reducing overconfidence leads to a reduction of the pull-to-center effect. Research in the BDG context has also explored the effects of overconfidence. Ancarani *et al.* (2016) show that decision makers are less cautious in ordering inventory when they are overconfident. They recommend that organizations be cognizant of this fact and manage expectations of the decision makers to mirror the real world. Feedback and guidance has been used in judgmental forecasting research to mitigate overconfidence (Arvan *et al.*, 2018; Perera *et al.*, 2019). Literature on inventory and ordering decisions will benefit from studies looking at how feedback and guidance may deter overconfidence.

Subject heterogeneity has emerged to the fore in recent years as researchers are keen to explore individual differences of decision makers, as opposed to the mean (Hyndman and Embrey, 2018). Differences in subject backgrounds or attributes in ordering decisions has been explored using behavioral experiments. Becker-Peth *et al.* (2018) attempts to understand the effect of subject heterogeneity in the newsvendor problem. More research is required on understanding the effect of subject heterogeneity on inventory and ordering decisions. Understanding the effect of factors such as education and age will greatly support the industry in determining the ideal profile of an ordering decision maker.

An age old debate has been on whether professional experience significantly improves ordering decisions. Croson and Donohue (2006) compare the results they obtained from undergraduate student participants in an inventory control experiment with results from industry professional participants. They find that results from the experienced professionals are consistent with those of the students. Bolton and Katok (2008) similarly find that the performance of experienced executive MBA students was consistent with undergraduate students. Bolton *et al.* (2012) formally investigate this issue in detail and find no evidence to suggest that professionals perform better than students in ordering decision tasks. Subsequent work has lent further support to this conclusion (Moritz *et al.*, 2013). As a result of these consistent findings, the criticism against laboratory experiments using student samples are ineffective in studying and predicting real-world behavior has been largely debunked. See Thomas (2011) for a detailed discussion.

With the globalization of supply chains, cultural influence has gained attention as a major determinant of decision-making. Thus, cultural differences in ordering decisions has gained popularity over the past decade (Cui *et al.*, 2013; Feng *et al.*, 2011; Özer *et al.*, 2014). The earliest such study repeats an experiment conducted in USA (Bolton and Katok, 2008) to find Chinese subjects displaying a stronger pull-to-center effect (Feng *et al.*, 2011). The authors explain this by using the “doctrine of the mean” which is ingrained in the Chinese culture and attributed to Confucius. The doctrine states that it is bad to go beyond a target as well as to fall short of it (Feng *et al.*, 2011). One experiment entailed to the subjects inquiring the experimenter for further information prior to making their ordering decisions (Cui *et al.*, 2013). The findings show that the Chinese subjects gather significantly more relevant information than their American counterparts. However, the overall performance does not show a significant

difference between the subject groups. Trust and trustworthiness has also been a major influence in the literature. [Ozer et al. \(2014\)](#) conduct a buyer-supplier game between the two subject groups in USA and China. All these studies have focused on USA and China. [Castañeda et al. \(2019\)](#) conduct a laboratory experiment in Uganda investigating the newsvendor behavior pertaining to perishable supplies. They report differences in scalability and risk-sharing in developing economies compared to developed ones. Future studies should explore the cultural influence of decision makers coming from other parts of the world.

The literature also contains a study on the effect of gender on a newsvendor setting in a high and low margin setting ([de Véricourt et al., 2013](#)). The findings show that opposing genders do not behave differently in a low margin setting. Yet, females tend to order significantly less in the high margin setting. The authors claim that this difference is mediated by the risk preferences of males and females. Further research reports that female subjects anchor more on mean demand than males despite finding no mediation effect in the newsvendor problem ([Becker-Peth et al., 2018](#)). There is also evidence of females offering higher service levels than men in the newsvendor task ([Villa and Castañeda, 2019](#)). They also inspect the effect of power in decision-making. It's worth investigating whether these findings hold when the cultural background varies since cultural differences may influence how individual differences manifest.

Other streams of behavioral OR has investigated the effect of visual presentation on decision-making ([Bendoly, 2016](#); [Willemain, 1989, 1991](#)). Research on inventory and ordering decisions is yet to enter this domain. Given that prior work suggests that visualization has considerable effects on the final decision ([Bendoly, 2016](#)), it is recommended for researchers to focus on this niche. In a similar vein, it would be interesting to observe how decision makers respond to changes in the currency in inventory and ordering decisions ([Li et al., 2019](#)). This would inform the community of the importance of context as most publications are tethered to either imaginary currencies or numbers that do not have an assigned currency.

6. Conclusion

Inventory ordering decisions are critical to effective supply chain management. While normative research has extensively investigated this topic, it is only recently that these decisions have been investigated through a behavioral lens. This paper provides the first systematical review of research on the topic that is conducted using behavioral experiments as the primary or complimentary research method. Our detailed examination yielded multiple findings of potential benefit to scholars and practitioners alike.

Our hope is that this review serves as an exhaustive yet convenient resource to researchers interested in the state of the experimental research on inventory replenishment. As our efforts showed, this topic has been increasing in prevalence in the literature over the last 20 years. By detailing the existing body of work, we illuminate two key factors. First, we catalog, to date, the work that has been done in this specific research space - in terms of topics and themes addressed, as well as influential studies and common experimental methods employed. In doing so, we hope that our efforts give interested researchers a clearer picture of a second factor, which is where potential exists to advance the topic in the years to come. Our work highlights several important questions that have yet to be explored, as well as the fact that, being saturated with laboratory experiments, the research space is ripe for insights derived from alternative methods. We expect that rich contributions can be made to the body of knowledge in this space through future studies complementing existing work with heretofore underutilized research approaches such as field experiments or managerial interviews.

The field of BOM at large is poised for continued growth, and for good reason – the impacts of human judgment and decision permeates every area of a firm and its supply chain.

As a result, our work stands to prove useful beyond academic settings by helping to guide managers in practice as they seek a better understanding of the most important asset of supply chains – the human decision makers involved in their operations.

Notes

1. Evidence supports the use of Scopus (managed by Elsevier publishing) as an excellent source of operations and supply chain peer-reviewed papers (e.g., [Fahimnia et al., 2015](#); [Fahimnia et al., 2015](#)). Furthermore, Scopus is the largest abstract and citation database for peer-reviewed academic literature (i.e., scientific journals, books and conference proceedings) within the fields of science, technology, medicine, social sciences, and arts and humanities. In fact, Scopus covers over 20,000 peer-reviewed journals, which is more comprehensive than the Web-of-Science database (managed by Thomson Reuters; includes only ISI indexed journals thus limiting to approximately 12,000). The only apparent drawback to Scopus is the limited access to papers in certain journals pre-1996.
2. Those who wish you learn more about the mechanics and capabilities of VOSviewer are referred to [Van Eck and Waltman \(2010, 2014\)](#).

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Table A1.
Salient characteristics
of the
experimental works

Appendix

Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
Sterman (1987)	Laboratory	Points		48 ^a		USA, Europe, Soviet Union	4	Capital investments for orders	
Sterman (1989)	Laboratory	Monetary		192 ^a		USA	2	BDG / 4 echelons	Misperception of feedback
Diehl and Sterman (1995)	Laboratory	Monetary	14	4		USA	3	Multi-period	Dynamic complexity and behavior
Schweitzer and Cachon (2000)	Laboratory	Monetary		78		USA	1	Newsvendor	Decision biases under known demand
Grosen and Donohue (2002)	Laboratory	N/A		Details aren't provided ^a		USA	2	BDG / different echelons	Synthesizing unpublished past experiments to find cognitive limitations
Crosen and Donohue (2003)	Laboratory	Monetary	84			USA	2	BDG / 4 echelons	Point of sales (POS) data sharing
Snijders <i>et al.</i> (2003)	Vignette	None	60		30	Netherlands	3, 4	Purchasing	Comparing computers, students and experts
Steckel <i>et al.</i> (2004)	Laboratory	Monetary		300		USA	2	BDG / 3 echelons	POS data sharing and cycle times
Tazelaar and Snijders (2004)	Vignette	None	60		30	Netherlands	3, 4	Purchasing	Comparing computers, students and experts in a team environment

(continued)

Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
Crosen and Donohue (2005)	Laboratory	Monetary		96		USA	2	BDG / 4 echelons	Availability of downstream inventory information Access to inventory information
Crosen and Donohue (2006)	Laboratory	Monetary	160			USA	2	BDG / 4 echelons	Access to inventory information
Wu and Katok (2006)	Laboratory	Monetary	192			USA	2	BDG / 4 echelons	Learning and communication
Bolton and Katok (2008)	Laboratory	Monetary		234 ^b		USA	1, 4	News vendor	Experience and feedback
Su (2008)	Laboratory	N/A		38 ^a		N/A	1, 4	News vendor	Testing logit choice model proposed in the paper
Katok <i>et al.</i> (2008)	Laboratory	Monetary (none for industry participants)	76		8	USA	3	Contracts and replenishments	Effect of service level targets and review periods
Benzion <i>et al.</i> (2008)	Laboratory	Monetary	60			Israel	1	News vendor	Decision-making under unknown demand
Hung and Ryu (2008)	Laboratory	N/A			82 ^c	N/A	2	BDG / 4 echelons	Effect of risk preference
Rong <i>et al.</i> (2008)	Laboratory	Monetary and course credits	84	8		USA	2	BDG / 4 echelons	Reverse bullwhip effect

(continued)

Table A1.

Table A1.

Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
Bostian <i>et al.</i> (2008)	Laboratory	Monetary	82			USA	1	News vendor	Adaptive learning model to explain pull-to-center effect and individual decisions
Niranjan <i>et al.</i> (2009)	Laboratory	Course credits		120		India	2	BDG / 4 echelons	Behavioral causes of the BWE
Cantor and MacDonald (2009)	Laboratory	Course credits	131			USA	2, 4	BDG / 4 echelons	Construal level theory and moderating effect of information
Delhoum and Scholz-Reiter (2009)	Laboratory	N/A		130 ^b		Germany	2	Supply net game	Inventory control decision patterns and BWE
Lurie and Swaminathan (2009)	Laboratory	Monetary and course credits	491			N/A	1	News vendor	Timeliness of feedback and information
Benzion <i>et al.</i> (2010)	Laboratory	Monetary	121			Israel	1	News vendor	Effects of known and unknown demand
Gavirneni and Isen (2010)	Verbal protocol analysis	Monetary		21		USA	1	News vendor	Anatomy of news vendor decision
Ho <i>et al.</i> (2010)	Laboratory	Monetary	160			USA	1, 4	Multilocation news vendor	Testing reference dependence
Kremer <i>et al.</i> (2010)	Laboratory	Monetary		53 ^b		Germany	1	News vendor	Effects of random errors

(continued)

Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
Feng <i>et al.</i> (2011)	Laboratory	Monetary	113 ^b			China and USA	1, 4	News vendor	Cross-border differences between USA and China
Tokar <i>et al.</i> (2011)	Laboratory	Monetary	30		76	USA	2	Multi-period and modified BDG	Retail promotions and inventory sharing for known demand
Niranjan <i>et al.</i> (2011)	Laboratory	Course credits		88		N/A	2	BDG / 4 echelons	Overordering behavior and SLU
Cantor and Katok (2012)	Laboratory	Monetary	106 ^b			USA	2	Two-tier serial supply chain	Seasonality and production smoothing
Tokar <i>et al.</i> (2012)	Laboratory	Course credits, monetary or none	195 ^b		36	USA	2	BDG / Single- tier and 4-tier	Learning and debiasing
Chen <i>et al.</i> (2012)	Laboratory	Monetary	60			China	4	Capacity allocation game	Retailer order behavior
Bolton <i>et al.</i> (2012)	Laboratory	Monetary	32	31	26	Germany	1, 4	News vendor	Comparing managers with students
Cui <i>et al.</i> (2013)	Verbal protocol analysis Laboratory	N/A		21		China and USA	1, 4	News vendor	Cultural differences in decision-making
de Véricourt <i>et al.</i> (2013)	Laboratory	Monetary		102	349	Singapore, USA and more	1, 4	News vendor	Gender differences in decision-making

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Table A1.

Table A1.

Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
Mortz <i>et al.</i> (2013)	Laboratory	Monetary and none	176 ^b		313	USA	1, 4	Newsvendor	Cognitive reflection theory
Chen <i>et al.</i> (2013)	Laboratory	Monetary	286 ^b			USA	1	Newsvendor	Effect of payment schemes on inventory
Ancarani <i>et al.</i> (2013)	Laboratory	Monetary and course credits		124		Italy	2	BDG / 4 echelons	Decisions Bullwhip effect under stochastic lead times
Wu (2013)	Laboratory	Monetary	45	53		USA	2	Two-tier SC	Repeated interactions on supply chain contracts
Strohhecker and Größler (2013)	Laboratory	Monetary	135			Germany	3, 4	Multi-period	Effect of personal traits intelligence, personality, interest and knowledge
Bloomfield and Kulp (2013)	Laboratory	Monetary		115 ^b		USA	1, 3	Multi-period and newsvendor	Effect of transit times and durability of products
Becker-Peth <i>et al.</i> (2013)	Laboratory	Monetary	96			Germany	1	Newsvendor	Contract parameters and ordering behavior
Ren and Croson (2013)	Laboratory	Monetary		243 ^b		USA	1, 4	Newsvendor	Overconfidence of newsvendors
Feiler <i>et al.</i> (2013)	Laboratory	Monetary and donation	108	141		USA	1	Newsvendor	Judgment in censored environments

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Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
Tokar <i>et al.</i> (2014)	Laboratory	Course credits	71	120		USA	3	Multi-period	Bracing behavior
Rudi and Drake (2014)	Laboratory	Lottery for iPod	269			Norway	1	Newsvendor	Level behavior, adjustments, observation bias
Ockenfels and Selten (2014)	Laboratory	Monetary		340 ^b		Germany	1, 4	Newsvendor	Impulse balancing
Ozer <i>et al.</i> (2014)	Laboratory	Monetary	186			USA and China	2, 4	Two-tier supply chains	Trust and trustworthiness, information sharing between countries/cultures
Ouyang (2014)	Laboratory	N/A		12		USA	2	BDG / 4 echelons	Advanced demand information
Kremer <i>et al.</i> (2014)	Laboratory	Monetary		152 ^b		USA	1	Newsvendor	Behavioral value of demand information
Wu and Chen (2014)	Laboratory	Monetary	126			USA	1	Newsvendor	Non-optimizing behavior
Kremer and Van Wassenhove (2014)	Laboratory	Monetary	76			USA	1	Newsvendor	Willingness to pay to avoid supply-demand mismatches and reduce inventory risk
Crosan <i>et al.</i> (2014)	Laboratory	Monetary	88			USA	2	BDG / 4 echelons	Coordination risk
Schiffels <i>et al.</i> (2011)	Laboratory	Monetary		148 ^b		Germany	1	Newsvendor	Assessment of costs

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Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
Gurnani <i>et al.</i> (2014)	Laboratory	Monetary		204 ^b		USA	1	Single-period	Supply selection under disruption
Lau <i>et al.</i> (2014)	Laboratory	Monetary		212 ^a		N/A	1	Newsvendor	Effect of heterogeneity on the pull-to-center effect
Chen and Zhao (2015)	Laboratory	Monetary	180			China	4	Capacity allocation games	Retailer behavior in demand uncertainty
Zhao and Zhao (2015)	Laboratory	Monetary	15	85		China	2	One decision- maker in a 5 echelon SC	Effects of information sharing
Käki <i>et al.</i> (2015)	Laboratory	Monetary		60 ^b	11	Finland	1	Newsvendor	Pull-to-center effect under supply uncertainty
Kocabiyikoglu <i>et al.</i> (2015)	Laboratory	Course credits		112 ^a		Turkey	1	Newsvendor and revenue management	Comparing decision-making with normative behavior
Hartwig <i>et al.</i> (2015)	Laboratory	N/A		96 ^a		Germany	2	Two-period and two-tier supply chains BDG / 4 echelons	Strategic inventory management
Sarkar and Kumar, (2015)	Laboratory	Monetary	200			USA	2		Supply chain disruption and information
Villa <i>et al.</i> (2015)	Laboratory	Monetary	80			Colombia	3	Multi-period	sharing Ordering decisions under delays

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Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
Narayanan and Moritz (2015)	Laboratory	Course credits	340	60		USA	2	BDG / 4 echelons	Cognitive reflection and supply line underweighting Phantom ordering
Sterman and Dogan (2015)	Laboratory	Monetary	88			USA	2	BDG / 4 echelons	Human behavior in the competing newsvendor problem
Ovchinnikov <i>et al.</i> (2015)	Laboratory	Monetary		88 ^b		USA	1	Competing newsvendor	Effects of framing on inventory decisions
Tokar <i>et al.</i> (2016)	Laboratory	Monetary and none	311	51	48	USA	3	Multi-period	Determining both the order quantity and the price
Kocabiyikoglu <i>et al.</i> (2016)	Laboratory	Monetary and course credits	189			Turkey	1	Price setting newsvendor	Information reliability
Spiliotopoulou <i>et al.</i> (2016)	Laboratory	Monetary		125 ^a		USA	1	Newsvendor	Relationship between sales and fill rates
Craig <i>et al.</i> (2016)	Field			513 SKUs		Germany	3	Multi-period	Overconfidence
Ancarani <i>et al.</i> (2016)	Laboratory	Monetary and course credits		152	24	Italy	2, 4	BDG / 4 echelons	Reference points in revenue sharing contracts in the newsvendor problem
Becker-Peth and Thonemann (2016)	Laboratory	Monetary	103 ^b			Germany	1	Newsvendor	

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Table A1.

Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
Zhao <i>et al.</i> (2016a)	Laboratory	Monetary		96 ^b		China	1	News vendor	Censored demand information (demand realization under unknown demand)
Zhao <i>et al.</i> (2016b)	Laboratory and neuroimaging	Monetary		81 ^b		China	4	Capacity allocation games	Elicitation methods (strategy and direct-response)
Zhao and Zhao (2016)	Laboratory	Monetary		80 ^b		China	1	Competing news vendor	Decision biases
Fügener <i>et al.</i> (2017)	Laboratory	Monetary	107		66	Germany	4	Capacity planning	Planning surgeries in operating theatres
Haines <i>et al.</i> (2017)	Laboratory	Gifts	120			USA	2	BDG / 4 echelons	Metacognitive perspective on decision making
Robinson <i>et al.</i> (2017)	Laboratory	N/A		7		UK	1	News vendor	Sample size problem
Stangl and Thonemann (2017)	Laboratory	Monetary		367 ^b		Germany	3	Inventory metrics	EOQ & mutually exclusive order decisions
Scheele <i>et al.</i> (2017)	Laboratory		64			Germany	1, 4	News vendor	Designing incentive schemes
Feng and Zhang (2017)	Laboratory	Monetary		263 ^a		China	1	Competing news vendor	Duopolistic ordering behavior
<i>(continued)</i>									

Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
D'Urso <i>et al.</i> (2017)	Laboratory	Course credits		106		Italy	1	News vendor	Anchoring and adjustment with and without demand information Task
Lee and Siensen (2017)	Laboratory	Monetary	196	42	27	USA	1, 4	News vendor	decomposition
Tong <i>et al.</i> (2018)	Laboratory	Monetary, gift cards and donations	353	78		USA	1	News vendor and forecasting	Censorship bias
Lee <i>et al.</i> (2018)	Laboratory	Monetary		474 ^a		USA	1	News vendor	Using MTurk on news vendor problem, procurement auction and chain contracts
Zhao and Zhao (2018)	Laboratory	Monetary		64 ^b		China	1	Competing news vendor	Incomplete information with the strategy method and the direct-response method
Villa and Castañeda (2018)	Laboratory	Monetary		252 ^b		Colombia	1	News vendor	Transshipments and news vendor interactions
Schultz <i>et al.</i> (2018)	Laboratory	Monetary	23	126		USA	1	News vendor	Changing risk preferences using framing

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Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
Castaneda and Gonçalves (2018)	Laboratory	Monetary	95			Colombia	1	News vendor	Newsstand problem
Zhang <i>et al.</i> (2018)	Laboratory	Monetary	215			N/A	1, 4	News vendor	Inventory decisions in the presence of strategic customers
Becker-Peth <i>et al.</i> (2018)	Laboratory	Monetary		220 ^b		Germany	1, 4	News vendor	Individual risk preference on order quantity
Castañeda <i>et al.</i> (2019)	Laboratory	Monetary	40			Uganda	1, 4	News vendor	Supply chain contracts for a news vendor problem in a developing economy
Kaboli <i>et al.</i> (2019)	Laboratory	Monetary		48 ^b		Switzerland	3	Multi-period/4 echelons	Replenishment behavior in sequential supply chains
Khan <i>et al.</i> (2019)	Laboratory	N/A		416 ^a		India	1	BDG / 4 echelons	Analyzing the Bullwhip effect
Li <i>et al.</i> (2019)	Laboratory	None			107	China	1, 4	News vendor	Individual and cultural differences
Quiroga <i>et al.</i> (2019)	Laboratory	Monetary		75 ^b		USA	1	Competing news vendor	Behavioral Ordering, Competition and Profits

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Author(s) (year)	Type of experiment	Form of incentive	Undergraduate	Participants Postgraduate	Professionals	Location of experiments	Cluster	Research focus	Additional details
Villa and Castaneda (2019)	Laboratory	Monetary	131			Colombia	1, 4	News vendor	Influence of power and gender
Zhang and Siensen (2019)	Laboratory	Monetary	200			USA	1, 4	News vendor	A meta-analysis of past news vendor experiments
Becker-Peth <i>et al.</i> (2020)	Laboratory	Monetary		126 ^b		Germany	3	Multi-period	Budget cycles in multi-period ordering
Feng and Gao (2020)	Laboratory	Monetary		167 ^b		China	1	News vendor	Testing optimal recommendations
Legend * - Participants for the news vendor experiment a - Includes undergraduates, postgraduates, and professionals b - Includes undergraduates and postgraduates c - Includes postgraduates and professionals N/A - Not Available									

Table A1.